# EXERCISE - O

## SINGLE CORRECT TYPE QUESTIONS

### *Introduction of Metallurgy, types of ores*

**1.** Which of the following is not an ore of aluminium?

(A) Bauxite (B) Corundum (C) Limonite (D) Kaolinite

**CMT102**

**2.** Which of the following set of metals are mostly found as sulphide ores :

(A) Zn, Cu, Mg (B) Zn, Cu, Pb (C) Fe, Al, Sn (D) Cu, Ag, Au

**CMT103**

**3.** Which mineral has been named incorrectly?

(A) Bauxite : $Al_2O_3.2H_2O$

(B) Corundum : $Al_2O_3$

(C) Cryolite : $3NaF.AlF_3$

(D) Fluorspar : $Be_3Al_2Si_6O_{18}$

**CMT104**

**4.** Select correct statement

(a) Aluminium is third most abundant element in earth crust.

(b) Aluminium is major component of many igneous minerals including mica and clays.

(c) Many gem stones are impure forms of $Al_2O_3$

(d) Iron is the second most abundant metal in the earth crust

(A) Only a, b and c (B) Only a & c (C) a, b, c and d (D) Only a, b & d

**CMT107**

**5.** The process of the isolation of a metal by dissolving the ore in aqueous solution of suitable chemical reagent followed by precipitation of the metal by a more electropositive metal is called:

(A) hydrometallurgy (B) electrometallurgy

(C) zone refining (D) electrorefining

**CMT012**

### *Metallurgical Processes*

**6.** Gravity separation process may be used for the concentration of :

(A) Copper pyrites (B) Bauxite (C) Haematite (D) Argentite

**CMT112**

**7.** Which of the following ores is concentrated by magnetic separation method ?

(A) Bauxite (B) Chromite (C) Argentite (D) Dolomite

**CMT114**

**8.** Which one of the following reactions represents a calcination reaction?

(A) $HgS + O_2 \rightarrow Hg + SO_2$ (B) $2CuFeS_2 + 4O_2 \rightarrow Cu_2S + 2FeO + 3SO_2$

(C) $CuCO_3.Cu(OH)_2 \rightarrow CuO + CO_2 + H_2O$ (D) $Al_2O_3.2H_2O + 2NaOH \rightarrow 2NaAlO_2 + 3H_2O$

**CMT119**

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**9.** Select incorrect statement regarding calcination?

(A) In calcination usually, the hydroxide ore and carbonate ore are converted to oxide.

(B) It is an endothermic process.

(C) It is carried out in absence of air.

(D) After calcination surface area of ore decreases.

**CMT121**

**10.** Incorrect statement regarding roasting is :

(A) Usually the sulphide ore is converted to oxide before reduction because oxides ore easier to reduce.

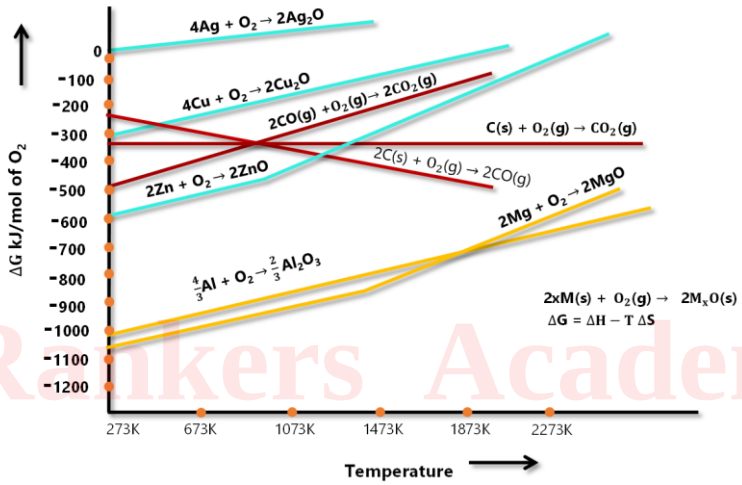
(B) It is an endothermic process.

(C) It is carried out in reverberatory furnace.

(D) Slag formation may take place during roasting.

**CMT123**

**11.** Considering Ellingham diagram, which of the following metals can be used to reduce alumina ?




(A) Fe (B) Cu (C) Mg (D) Zn

**CMT125**

**12.** Which of the following metals cannot be extracted by the carbon reduction process?

(A) Zn (B) Fe (C) Al (D) Sn

**CMT127**

**13.** Carbon cannot be used in the reduction of $Al_2O_3$ because :

(A) It is non-metal

(B) The heat of formation of $CO_2$ is more than that of $Al_2O_3$

(C) Pure carbon is not easily available

(D) The heat of formation of $Al_2O_3$ is too high

**CMT129**

**14.** Ellingham diagram represents:

(A) Change of $\Delta G^\circ$ with temperature. (B) Change of $\Delta H^\circ$ with temperature.

(C) Change of $\Delta G^\circ$ with pressure. (D) Change of $(\Delta G^\circ - T\Delta S^\circ)$ with temperature.

**CMT130**

**15.** Which of the following metals is obtained by the self reduction process?

(A) Copper (B) Tin (C) Gold (D) Magnesium

**CMT132**



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**16.** Identify the metal M whose extraction is based on the following reactions :

$MS + 2O_2 \rightarrow MSO_4$

$2MS + 3O_2 \rightarrow 2MO + 2SO_2$

$MS + 2MO \rightarrow 3M + SO_2$

$MS + MSO_4 \rightarrow 2M + 2SO_2$

(A) Zinc (B) Aluminium (C) Lead (D) Silver

**CMT134**

**17.** In which of the following pair of metals, both are commercially extracted from their respective ores by self reduction method ?

(A) Zn, Cu (B) Pb, Cu (C) Sn, Zn (D) Al, Ag

**CMT136**

**18.** Poling process :

(A) Reduces $SnO_2$ to Sn.

(B) Involves the liberation of reducing gases like hydrocarbons.

(C) Uses poles of freshly cut green wood.

(D) All of the above are correct.

**CMT138**

**19.** The method of zone refining of metals is used for the purification of :

(A) Zn (B) Al (C) Cu (D) None

**CMT140**

**20.** In the electrolytic refining of lead, Sb, Cu, Ag and Au are found :

(A) On anode (B) In electrolyte solution

(C) In anode mud (D) In cathode mud

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**CMT144**

**21.** Van Arkel method of purification of metals involves converting the metal to:

(A) Volatile stable compound. (B) Volatile unstable compound.

(C) Non-volatile stable compound. (D) None of these.

**CMT148**

**22.** The iron obtained from the blast furnace is called :

(A) Pig iron (B) Cast iron (C) Wrought iron (D) Steel

**CMT167**

**23.** Process used for extraction of Ag from its native ore is :

(A) Self reduction (B) Carbon reduction

(C) Leaching and Hydrometallurgy (D) Electrolytic reduction

**CMT181**

**24.** Which of the following metals cannot be extracted by the carbon reduction process ?

(A) Zn (B) Fe (C) Al (D) Sn

**CMT002**

**25.** In Ellingham diagram which of the following reaction gives line almost parallel to x-axis ?

(A) $CO(g) + \frac{1}{2} O_2(g) \rightarrow CO_2(g)$ (B) $C(s) + O_2(g) \longrightarrow CO_2(g)$

(C) $C(s) + \frac{1}{2} O_2(g) \longrightarrow CO(g)$ (D) $2Al(s) + \frac{1}{2} O_2(g) \longrightarrow Al_2O_3(s)$

**CMT005**



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**26.** An ore of metal 'M' (After crushing and grinding) $\xrightarrow{\text{Roasting}}$ Metal oxide $\xrightarrow[\text{Carbon}]{\text{Reduction}}$ 'M'

'M' may be :-

(A) Al (B) Zn (C) Hg (D) All of these

**CMT007**

**27.** Which of the following cannot be reduced by carbon ?

(A) PbO (B) $SnO_2$ (C) ZnO (D) MgO

**CMT008**

**28.** The following equation represents a method of purification of nickel by :-

$Ni\ (impure) + 4CO \xrightarrow{320K} Ni(CO)4 \xrightarrow{420K} Ni\ (pure) + 4CO$

(A) Cupellation (B) Mond's process

(C) Van Arkel method (D) Zone refining

**CMT010**

**29.** Refractory materials are generally used in furnaces because

(A) they are chemically inert (B) they can withstand high temperature

(C) they do not contain impurities (D) they decrease melting point of ore

**CMT019**

**30.** Which of the following statements is correct regarding the slag formation during the extraction of a metal like copper or iron.

(A) The slag is lighter and has lower melting temperature than the metal

(B) The slag is heavier and has lower melting temperature than the metal

(C) The slag is lighter and has higher melting temperature than the metal

(D) The slag is heavier and has higher melting temperature than the metal

**CMT020**

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### *Extraction of metals*

**31.** Refining of Gold is done by :

(A) Parke's process (B) Parting process (C) Polling process (D) Kroll's process

**CMT096**

**32.** Extraction of gold and silver involves leaching with $CN^{\ominus}$ ion. Silver is later recovered by :

(A) Liquation (B) Distillation (C) Zone refining (D) Displacement with Zn

**CMT118**

**33.** To purify the 'blister copper' it is melted in a furnace and then stirred with the poles of freshly cut green wood. The purpose of this is :

(A) To remove the gases dissolved in blister copper.

(B) To bring the impurities to the surface of the molten liquid.

(C) To increase the carbon percentage.

(D) To reduce the metallic oxide to metal by hydrocarbons liberated by green wood

**CMT142**



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**34.** In electrolytic refining of lead, electrolyte consists of :

(A) $H_2SiF_6$ only (B) $PbSiF_6$ only (C) $H_2SiF_6$ only (D) $H_2SiF_6$ and $PbSiF_6$

**CMT146**

**35.** In electrolysis of $Al_2O_3$ by Hall-Heroult process :

(A) Cryolite $Na_3[AlF_6]$ lowers the melting point of $Al_2O_3$ and increases its electrical conductivity.

(B) Al is obtained at cathode and probably $CO_2$ at anode.

(C) Oxygen liberated at anode reacts with carbon anode liberating CO and $CO_2$.

(D) All of the above are correct.

**CMT152**

**36.** In the electrolytic reduction of fused mixture of $Al_2O_3$ and $Na_3AlF_6$ gas liberated at graphite anode is

(A) $F_2$ (B) $OF_2$ (C) $O_2$ (D) $O_3$

**CMT154**

**37.** Aluminium metal is purified by :

(A) Hoope's electrolytic reduction process (B) Dow's process

(C) Serpeck's process (D) Baeyer's process

**CMT156**

**38.** Roasting of ZnS is done at high temperature and in controlled air current to -

(A) Prevent formation of ZnO (B) Prevent formation of $ZnSO_4$

(C) Prevent formation of Zn (D) None of these

**CMT158**

**39.** Blister copper is :

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(A) Impure copper.

(B) Obtained in self-reduction process during bessemerisation.

(C) Both (A) and (B) are correct

(D) None of the above

**CMT160**

**40.** Which of the following statement is incorrect about the extractive metallurgy of copper?

(A) Matte chiefly consists of iron sulphide and some ferrous oxide

(B) The impurity of iron oxide is removed as fusible slag, $FeSiO_3$ during roasting

(C) The copper pyrite is concentrated by froth floatation process

(D) Copper is obtained by self-reduction in Bessemer converter

**CMT162**

**41.** Identify M and N in the following reaction :

$$\text{Copper glance} \xrightarrow{\text{Controlled heating in presence of air}} M + SO_2$$

$$M \xrightarrow{N} Cu + SO_2 (\uparrow)$$

(A) $M = Cu_2O + Cu_2S$, N = only heating

(B) $M = Cu_2O$, N = Carbon reduction

(C) $M = Cu_2O$, N = Electrolytic reduction

(D) None of these

**CMT163**



For More Join: @IITJEE_Advanced

**42.** If CaO (flux) is added in excess during smelting of cassiterite, then :

(A) Calcium stannite will form
(B) Ferrous silicate will not form
(C) Calcium silicate will not form
(D) None of these

**CMT171**

**43.** Main source of lead is galena. It is converted to Pb by :

(X) : $PbS \xrightarrow[\Delta]{air} PbO + SO_2$, $PbO \xrightarrow{C} Pb + CO_2$

(Y) : $PbS \xrightarrow[\Delta]{air} PbO + PbS \longrightarrow Pb + SO_2$

(Z) : $PbS \xrightarrow[\Delta]{air} PbO + SO_2$, $PbO \xrightarrow{CO} Pb + CO_2$

Self - reduction process is :

(A) X (B) Y (C) Z (D) none

**CMT173**

**44.** When impurity content is high then lead is extracted from galena by?

(A) Self reduction
(B) Carbon reduction
(C) Electrolytic reduction
(D) Hydrometallurgy

**CMT175**

**45.** Process used for extraction of Mg from molten $MgCl_2$ is :

(A) Self reduction
(B) Carbon reduction
(C) Electrolytic reduction
(D) Hydrometallurgy

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**CMT177**

**46.** In Dow's process, Sea water is treated with:

(A) First by Coke then NaOH
(B) First by NaCl then Coke
(C) First by $Ca(OH)_2$ then HCl
(D) First by $SiO_2$ then Coke

**CMT179**

**47.** Lead present in silver is removed by :

(A) Cupellation process
(B) Parke's process
(C) Mond's process
(D) Van Arkel process

**CMT183**

**48.** Silver is removed from impure gold by:

(A) Cupellation
(B) Heating with borax
(C) Heating with conc. $H_2SO_4$
(D) Heating with Coke

**CMT185**

**49.** Blister copper is :

(A) Impure copper.
(B) Obtained in self reduction process during bessemerization.
(C) Both (A) and (B) are correct.
(D) None is correct.

**CMT001**



For More Join: @IITJEE_Advanced

**50.** During the electrolytic reduction of aluminium, the carbon anodes are replaced from time to time because:
(A) the carbon anodes get decayed
(B) the carbon prevents atmospheric oxygen from coming in contact with aluminium
(C) oxygen liberated at the carbon anodes reacts with anodes to form CO and $CO_2$
(D) carbon converts $Al_2O_3$ to Al

**CMT003**

**51.** Silver can be separated from lead by :
(A) fractional crystallisation (B) liquation
(C) cupellation (D) addition of zinc (Parke's method)

**CMT004**

**52.** Which of the following is correct regarding extraction of Zn.
(A) Reduction of ZnO by CO is favourable at low temp.
(B) Zn is obtained as molten layer with slag.
(C) Reduction of ZnO by CO is reversible and high temperature is required.
(D) In reduction process $CO_2$ is the only gaseous product.

**CMT006**

**53.** In the Down's process of obtaining sodium, $CaCl_2$ is added -
(A) To get high yield of sodium (B) To improve the quality of sodium obtained
(C) To lower the operating temperature (D) To act as a reducing agent

**CMT009**

**54.** $Ag_2S + NaCN + Zn \longrightarrow Ag$
This method of extraction of Ag by complex formation and then its displacement is called:
(A) Parke's method (B) McArthur-Forest method
(C) Serpeck method (D) Hall's method

**CMT011**

**55.** Blister copper is refined by stirring molten impure metal with green logs of wood because such a wood liberates hydrocarbon gases (like $CH_4$). This process X is called _____ and the metal contains impurities of Y is _____.
(A) X = cupellation, Y = $CuO_2$ (B) X = poling, Y = $Cu_2O$
(C) X = poling, Y = CuO (D) X = cupellation, Y = CuO

**CMT013**

**56.** A piece of steel is heated until redness and then plunged into cold water or oil. This treatment of steel makes it
(A) soft and malleable (B) hard but not brittle
(C) more brittle (D) hard and brittle

**CMT014**

**57.** Modern method of steel manufacturing is
(A) open hearth process (B) L.D. Process
(C) Bessemerisation (D) Cupellation

**CMT015**

**58.** During electrolytic reduction of alumina, two auxiliary electrolytes X and Y are added to increase the electrical conductance and lower the temperature of melt in order to making fused mixture very conducting. X and Y are
(A) cryolite and fluorspar (B) cryolite and alum
(C) alum and fluorspar (D) fluorspar and bauxite

**CMT016**

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**59.** Railway wagon axles are made by heating iron rods embedded in charcoal powder. This process is known as:
(A) Sherardising (B) Annealing (C) Tempering (D) Case hardening

**CMT017**

**60.** Which of the following statement is correct regarding Cu-extraction
(A) In the smelting step carbon reduction takes places
(B) During partial roasting $Cu_2S$ remains almost unaffected
(C) In Bessemer converter, only self-reduction occur, not slag formation
(D) Blister forms in the blister Cu is due to dissolved $CO_2$

**CMT018**

## MULTIPLE CORRECT TYPE QUESTIONS

### *Metallurgical processes*

**61.** Roasting of copper pyrites is done :
(A) To remove moisture
(B) To oxidise free Sulphur and Antimony
(C) To convert pyrites completely into $Cu_2O$ and FeO
(D) To remove volatile organic impurities

**CMT168**

## MATRIX MATCH TYPE QUESTION

### *Introduction of Metallurgy types of ores*

**62.** Match Column-I with Column-II and select the correct answer using the codes given below :

| **Column-I (Metals)** | **Column-II (Ores)** |
|---|---|
| (1) Tin | (p) Zincite |
| (2) Zinc | (q) Cassiterite |
| (3) Copper | (r) Cerussite |
| (4) Lead | (s) Cuprite |

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Codes :

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| (A) | p | q | r | s |
| (B) | q | p | s | r |
| (C) | s | r | q | p |
| (D) | q | p | r | s |

**CMT109**

### *Metallurgical processes*

**63.** Match the method of concentration of the ore in column I with the ore in column II and select the correct alternate :

| **(I)** | **(II)** |
|---|---|
| (1) Magnetic separation | (A) $Cu_2S$ |
| (2) Froth floatation | (B) $Fe_2O_3$ |
| (3) Gravity separation | (C) $SnO_2$ |

| | 1 | 2 | 3 |
|---|---|---|---|
| (A) | A | B | C |
| (B) | B | A | C |
| (C) | C | A | B |
| (D) | B | C | A |

**CMT116**



For More Join: @IITJEE_Advanced

**64.** Match **Column-I** with **Column-II**

| **Column-I (Metals)** | | **Column-II (Method used for refining)** | |
|---|---|---|---|
| (A) | Iron & copper | (P) | Poling |
| (B) | Zirconium & Titanium | (Q) | Bessemerisation |
| (C) | Lead & Tin | (R) | Van-Arkel |
| (D) | Copper & Tin | (S) | Liquation |

**CMT187**

*Miscellaneous/Mixed*

**65.**

| **Metal** | **Purification Method** |
|---|---|
| (a) Cu | (i) Van arkel process |
| (b) Ga | (ii) Mond process |
| (c) Ni | (iii) Electrolytic refining |
| (d) Zr | (iv) Zone refining method |

(A) a – iii, b – iv, c – ii, d – i
(B) a – ii, b – iii, c – iv, d – i
(C) a – i, b – ii, c – iii, d – iv
(D) a – iii, b – iv, c – i, d – ii

**CMT150**

**66.** Match the following choosing one item from column X and the appropriate item from column Y.

| | **Column -X** | | **Column-Y** |
|---|---|---|---|
| (A) | Zinc from $ZnCO_3$ | (P) | Calcination |
| (B) | Lead from PbS | (Q) | Removal of iron |
| (C) | Cu from $CuFeS_2$ | (R) | Froth floatation process |
| (D) | Tin from cassiterite | (S) | Poling |

**CMT188**

**67.** Match column - I and column - II and select the correct answer using the codes given below the lists:

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| | **Column - I** | | **Column - II** |
|---|---|---|---|
| (A) | Cyanide process | (P) | Ultrapure Ge |
| (B) | Floatation process | (Q) | Dressing of HgS |
| (C) | Electrolytic reduction | (R) | Extraction of Al |
| (D) | Zone refining | (S) | Extraction of Au |

**CMT189**

**68.** Match the items of Column I with items of Column II and assign the correct code :

| | **Column I** | | **Column II** |
|---|---|---|---|
| (P) | Blistered Cu | (1) | Aluminium |
| (Q) | Blast furnace | (2) | $2Cu_2O + Cu_2S \rightarrow 6Cu + SO_2$ |
| (R) | Reverberatory furnace | (3) | Iron |
| (S) | Hall-Heroult process | (4) | $FeO + SiO_2 \rightarrow FeSiO_3$ |
| | | (5) | $2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$ |

**Code :**

(A) P → (2) ; Q → (3) ; R → (4) ; S → (1)
(B) P → (1) ; Q → (2) ; R → (3) ; S → (5)
(C) P → (5) ; Q → (4) ; R → (3) ; S → (2)
(D) P → (4) ; Q → (5) ; R → (3) ; S → (2)

**CMT190**



For More Join: @IITJEE_Advanced

# EXERCISE - S

## *Introduction of Metallurgy, types of ores*

**1.** In extraction of metal how many of the following ores calcination process is involved.
(i) Dolomite (ii) Malachite (iii) Calcite (iv) Copper pyrites
(v) Sylvine (vi) Cryolite (vii) Siderite (viii) Iron pyrite
(ix) Argentite
**CMT021**

**2.** How many of the following minerals containing Mg.
Magnetite, Carnallite, Epsom salt, Siderite **CMT028**

**3.** Find out the number of minerals given below contain iron as Fe(II).
Haematite, Magnetite, Limonite, Siderite, Chromite, Wolframite
**CMT029**

**4.** How many of the following ores of silver ?
Hornsilver, Cerrusite, Chalcopyrite, Galena, Anglesite, Argentite **CMT030**

**5.** Amongst the following, total number of sulphide ores are
Calamine, Sphalertie, Copper pyrites, Copper glance, Iron pyrites, Bauxite
**CMT191**

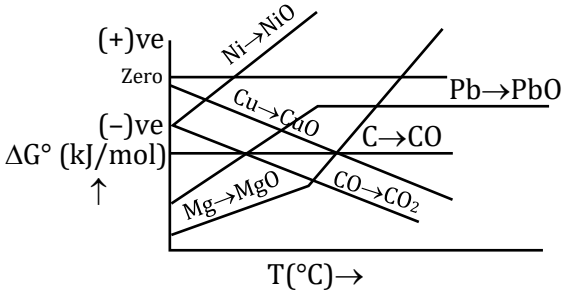
## *Metallurgical processes*

**6.** How many of the following metallurgies given below involve leaching ?
$Al_2O_3 \longrightarrow Al$, ; $Ag_2S \longrightarrow Ag$ ; Au(ore) $\longrightarrow$ Au ; $CuFeS_2 \longrightarrow Cu$ ; $PbS \longrightarrow Pb$
$MgCl_2 \longrightarrow Mg$ ; $FeCO_3 \longrightarrow Fe$ ; Low grade copper ore $\longrightarrow$ Cu ; $HgS \longrightarrow Hg$
**CMT022**

**7.** How many of these metals can be purified by vapour phase refining Ni, Ti, Zr, Ge, In, Fe, Cu, Au.
**CMT025**

**8.** Find the number of ore which are concentrated by magnetic separation method.
Cassiterite, Copper Glance, Chromite, Cinnabar **CMT026**

**9.** Find the number of curves which are wrongly presented in the Ellingham diagram.




**CMT027**

**10.** Find the number of metals which are commercially extracted by carbon reduction method
Pb, Fe, Zn, Mg, Al, Na, Au, Ag **CMT192**

**11.** Amongst the following ores, the total number of oxide ores are
Siderite, Magnetite, Hematite, Malachite, Zincite, Cuprite **CMT193**

## *Miscellaneous/Mixed*

**12.** Find the total number of process which involve formation of $Cl_2$ as one of the product/by product
- Down's cell
- Dow's sea- water process
- Deacons process
- Hall Heroult cell
- Castner-Kellner cell
- Bayer's process
- Ostwald process

**CMT023**

**13.** The number of following pairs is correctly matched
(i) Van Arkel method – Zirconium (ii) Mond Process – Titanium
(iii) Froth Floatation Method – Cerussite (iv) Distillation method – Zinc
(v) Poling Process – Copper (vi) Mac Aurther process – Gold **CMT024**



For More Join: @IITJEE_Advanced

# EXERCISE - JEE (Main) PYQ

**1.** The substance used as froth stabilisers in froth-floatation process is :

**[JEE (Main) 2012]**

(1) Copper sulphate (2) Aniline

(3) Sodium cyanide (4) Potassium ethyl xanthate

**CMT031**

**2.** Which of the oxide groups among the following cannot be reduced by carbon:-

**[JEE (Main) 2012]**

(1) $Fe_3O_4$, ZnO (2) PbO, $Fe_3O_4$ (3) $Cu_2O$, $SnO_2$ (4) CaO, $K_2O$

**CMT032**

**3.** In Goldschmidt alumino thermite process which of the following reducing agents is used :

**[JEE (Main) 2013]**

(1) Calcium (2) Coke (3) Sodium (4) Al-powder

**CMT033**

**4.** Calcination is the process in which : **[JEE (Main) 2013]**

(1) Ore is heated strongly below its melting point in the presence of excess of air and is used for the conversion of carbonates and hydrated oxide ores to their respective oxides.

(2) Ore is heated strongly below its melting point in the absence or limited supply of air and is used for conversion of sulphide ores to their respective oxides

(3) Ore is heated strongly below its melting point either in the limited supply or absence of air and is used to convert carbonates and hydrated oxide ores to their respective oxides

(4) Ore is heated strongly above its melting point in the limited supply of air to convert sulphide ores to their respective oxides.

**CMT034**

**5.** The metal that cannot be obtained by electrolysis of an aqueous solution of its salts is :

**[JEE (Main) 2014]**

(1) Cu (2) Cr (3) Ag (4) Ca

**CMT035**

**6.** The form of iron obtained from blast furnace is : **[JEE (Main) 2014]**

(1) Steel (2) Wrought Iron (3) Cast Iron (4) Pig iron

**CMT036**

**7.** In the context of the Hall-Heroult process for the extraction of Al, which of the following statements is false ? **[JEE (Main) 2015]**

(1) $Al^{3\oplus}$ is reduced at the cathode to form Al

(2) $Na_3AlF_6$ serves as the electrolyte

(3) CO and $CO_2$ are produced in this process

(4) $Al_2O_3$ is mixed with $CaF_2$ which lowers the melting point of the mixture and brings conductivity

**CMT037**



For More Join: @IITJEE_Advanced

**8.** Which one of the following ores is best concentrated by froth floatation method?

**[JEE (Main) 2016]**

(1) Malachite (2) Magnetite (3) Siderite (4) Galena

**CMT038**

**9.** What will occur if a block of copper metal is dropped into a beaker containing a solution of 1M $ZnSO_4$ ?

**[JEE (Main) 2016]**

(1) The copper metal will dissolve and zinc metal will be deposited

(2) No reaction will occur

(3) The copper metal will dissolve with evolution of oxygen gas

(4) The copper metal will dissolve with evolution of hydrogen gas

**CMT039**

**10.** In the leaching method, bauxite ore is digested with a concentrated solution of NaOH that produces 'X'. When $CO_2$ gas is passed through the aqueous solution of 'X', a hydrated compound 'Y' is precipitated. 'X' and 'Y' respectively are :- **[JEE (Main) 2018]**

(1) $Na[Al(OH)_4]$ and $Al_2O_3 \cdot x\ H_2O$ (2) $Al(OH)_3$ and $Al_2O_3 \cdot x\ H_2O$

(3) $Na[Al(OH)_4]$ and $Al_2(CO_3)_3 \cdot x\ H_2O$ (4) $Na\ AlO_2$ and $Al_2(CO_3)_3 \cdot x\ H_2O$

**CMT040**

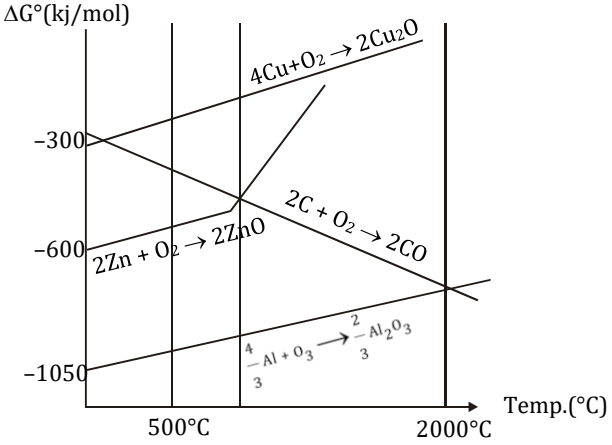
**11.** When metal 'M' is treated with NaOH, a white gelatinous precipitate 'X' is obtained, which is soluble in excess of NaOH. Compound 'X' when heated strongly gives an oxide which is used in chromatography as an adsorbent. The metal 'M' is **[JEE (Main) 2018]**

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(1) Ca (2) Al (3) Fe (4) Zn

**CMT041**

**12.** The correct statement regarding the given Ellingham diagram is: **[JEE (Main) 2019]**




(1) At 800°C, Cu can be used for the extraction of Zn from ZnO

(2) At 500°C, coke can be used for the extraction of Zn from ZnO

(3) Coke cannot be used for the extraction of Cu from $Cu_2O$.

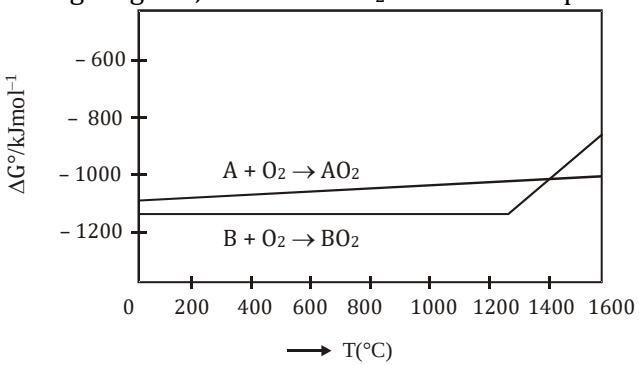
(4) At 1400°C, Al can be used for the extraction of Zn from ZnO

**CMT042**



For More Join: @IITJEE_Advanced

**13.** According to the following diagram, A reduces $BO_2$ when the temperature is : **[JEE (Main) 2020]**




(1) < 1400 °C (2) > 1400 °C (3) < 1200 °C (4) > 1200 °C but < 1400 °C

**CMT043**

**14.** The processes of calcination and roasting in metallurgical industries, respectively, can lead to :-

(1) Global warming and acid rain **[JEE (Main) 2020]**

(2) Photochemical smog and ozone layer depletion

(3) Global warming and photochemical smog

(4) Photochemical smog and global warming

**CMT044**

**15.** Which of the following ore is concentrated using group 1 cyanide salt ? **[JEE (Main) 2021]**

(1) Sphalerite (2) Calamine (3) Siderite (4) Malachite

**CMT045**

**16.** Match List-I with List-II : **[JEE (Main) 2021]**

| **List-I** Industrial process | **List-II** Application |
|---|---|
| (a) Haber's process | (i) $HNO_3$ synthesis |
| (b) Ostwald's process | (ii) Aluminium extraction |
| (c) Contact process | (iii) $NH_3$ synthesis |
| (d) Hall-Heroult process | (iv) $H_2SO_4$ synthesis |

Choose the correct answer from the options given below :

(1) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i) (2) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(3) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii) (4) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

**CMT046**

**17.** The metal that can be purified economically by fractional distillation method is: **[JEE (Main) 2021]**

(1) Fe (2) Zn (3) Cu (4) Ni

**CMT047**

**18.** Given below are two statements.

**Statement I:** The choice of reducing agents for metals extraction can be made by using Ellingham diagram, a plot of $\Delta G$ vs temperature.

**Statement II:** The value of $\Delta S$ increases from left to right in Ellingham diagram.

In the light of the above statements, choose the **most appropriate** answer from the options given below: **[JEE (Main) 2021]**

(1) Both **Statement I** and **Statement II** are true

(2) **Statement I** is false but **Statement II** is true

(3) Both **Statement I** and **Statement II** are false

(4) **Statement I** is true but **Statement II** is false

**CMT048**



For More Join: @IITJEE_Advanced

**19.** Match List – I with List – II **[JEE (Main) 2022]**

| | List - I | | List - II |
|---|---|---|---|
| (A) | Sphalerite | (I) | $FeCO_3$ |
| (B) | Calamine | (II) | PbS |
| (C) | Galena | (III) | $ZnCO_3$ |
| (D) | Siderite | (IV) | ZnS |

Choose the most appropriate answer from the options given below:

(1) (A) - (IV), (B) – (III), (C) – (II), (D) – (I) (2) (A) - (IV), (B) – (I), (C) – (II), (D) – (III)

(3) (A) - (II), (B) – (III), (C) – (I), (D) – (IV) (4) (A) - (III), (B) – (IV), (C) – (II), (D) – (I)

**CMT049**

**20.** The compound(s) that is(are) removed as slag during the extraction of copper is : **[JEE (Main) 2022]**

(1) CaO (2) FeO (3) $Al_2O_3$ (4) ZnO

(5) NiO

Choose the correct answer from the options given below :

(1) (3) (4) Only (2) (1), (2), (5) Only

(3) (1), (2) Only (4) (2) Only

**CMT050**

**21.** Match List I with List II. **[JEE (Main) 2023]**

| List – I | | List – II | |
|---|---|---|---|
| A. | Reverberatory furnace | I. | Pig Iron |
| B. | Electrolytic cell | II. | Aluminium |
| C. | Blast furnace | III. | Silicon |
| D. | Zone Refining furnace | IV. | Copper |

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(1) A – IV, B – II, C – I, D – III (2) A – I, B – IV, C – II, D – III

(3) A – I, B – III, C – II, D – IV (4) A – III, B – IV, C – I, D – II

**CMT051**



For More Join: @IITJEE_Advanced

# EXERCISE - JEE (Advanced) PYQ

**1.** In extractive metallurgy of zinc partial fusion of ZnO with coke is called _____ and reduction of the ore to the molten metal is called _____ (smelting, calcining, roasting, sintering). **[JEE (Advanced) 1988]**

**CMT072**

**2.** The methods chiefly used for the extraction of lead and tin from their ores are respectively : **[JEE (Advanced) 2004]**

(A) self reduction and carbon reduction
(B) self reduction and electrolytic reduction
(C) carbon reduction and self reduction
(D) cyanide process and carbon reduction

**CMT073**

**3.** Which ore contains both iron and copper? **[JEE (Advanced) 2004]**

(A) Cuprite (B) Chalcocite (C) Chalcopyrite (D) Malachite

**CMT074**

**4.** Match the extraction processes listed in column I with metals listed in column II. **[JEE (Advanced) 2006]**

| **Column I** | **Column II** |
|---|---|
| (A) Self reduction | (P) Lead |
| (B) Carbon reduction | (Q) Silver |
| (C) Complex formation and displacement by metal | (R) Copper |
| (D) Hall Heroult process | (S) Aluminium |

(A) ((A)-P,Q ; (B)-R ; (C)-Q ; (D)-S)
(B) ((A)-P,R ; (B)-P ; (C)-Q ; (D)-S)
(C) ((A)-R,S ; (B)-P ; (C)-Q ; (D)-P)
(D) ((A)-P,Q ; (B)-R ; (C)-Q ; (D)-P)

**CMT075**

**5.** Extraction for zinc from zinc blende is achieved by : **[JEE (Advanced) 2007]**

(A) electrolytic reduction
(B) roasting followed by reduction with carbon
(C) roasting followed by reduction with another metal
(D) roasting followed by self-reduction

**CMT076**

**6.** Match the conversions in **Column I** with the type(s) of reaction(s) given in **Column II**. Indicate your answer by darkening the appropriate bubbles of the 4 × 4 matrix given in the ORS. **[JEE (Advanced) 2008]**

| **Column I** | **Column II** |
|---|---|
| (A) $PbS \rightarrow PbO$ | (P) Roasting |
| (B) $CaCO_3 \rightarrow CaO$ | (Q) Calcination |
| (C) $ZnS \rightarrow Zn$ | (R) Carbon reduction |
| (D) $Cu_2S \rightarrow Cu$ | (S) Self reduction |

(A) (A)-P,S ; (B)-S ; (C)-Q,R ; (D)-P
(B) (A)-R ; (B)-Q ; (C)-P,; (D)-P,R
(C) (A)-P ; (B)-S ; (C)-R ; (D)-P
(D) (A)-P ; (B)-Q ; (C)-P,R ; (D)-P,S

**CMT077**

**7.** Native silver metal forms a water soluble complex with a dilute aqueous solution of NaCN in the presence of :- **[JEE (Advanced) 2008]**

(A) nitrogen (B) oxygen (C) carbon dioxide (D) argon

**CMT078**

**Paragraph for questions 8 to 10**

Copper is the most noble of the first row transition metals and occurs in small deposits in several countries. Ores of copper include chalcanthite ($CuSO_4 \cdot 5H_2O$), atacamite ($Cu_2Cl(OH)_3$), cuprite ($Cu_2O$), copper glance ($Cu_2S$) and malachite ($Cu_2(OH)_2CO_3$). However, 80% of the world copper production comes from the ore chalcopyrite ($CuFeS_2$). The extraction of copper from chalcopyrite involves partial roasting, removal of iron and self-reduction. **[JEE (Advanced) 2010]**



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**8.** Partial roasting of chalcopyrite produces :-
(A) $Cu_2S$ and FeO (B) $Cu_2O$ and FeO (C) CuS and $Fe_2O_3$ (D) $Cu_2O$ and $Fe_2O_3$
**CMT079**

**9.** Iron is removed from chalcopyrite as :- **[JEE (Advanced) 2010]**
(A) FeO (B) FeS (C) $Fe_2O_3$ (D) $FeSiO_3$
**CMT080**

**10.** In self-reduction, the reducing species is :- **[JEE (Advanced) 2010]**
(A) S (B) $O^{2\ominus}$ (C) $S^{2\ominus}$ (D) $SO_2$
**CMT081**

**11.** Extraction of metal from the ore cassiterite involves **[JEE (Advanced) 2011]**
(A) carbon reduction of an oxide ore (B) self-reduction of a sulphide ore
(C) removal of copper impurity (D) removal of iron impurity
**CMT082**

**12.** Oxidation states of the metal in the minerals haematite and magnetite, respectively, are **[JEE (Advanced) 2011]**
(A) II, III in haematite and III in magnetite (B) II, III in haematite and II in magnetite
(C) II in haematite and II, III in magnetite (D) III in haematite and II, III in magnetite
**CMT083**

**13.** In the cyanide extraction process of silver from argentite ore, the oxidizing and reducing agents used are : **[JEE (Advanced) 2012]**
(A) $O_2$ and CO respectively. (B) $O_2$ and Zn dust respectively.
(C) $HNO_3$ and Zn dust respectively. (D) $HNO_3$ and CO respectively.
**CMT084**

**14.** Sulphide ores are common for the metals - **[JEE (Advanced) 2013]**
(A) Ag, Cu and Pb (B) Ag, Cu and Sn (C) Ag, Mg and Pb (D) Al, Cu and Pb
**CMT085**

**15.** The carbon-based reduction method is **NOT** used for the extraction of **[JEE (Advanced) 2013]**
(A) tin from $SnO_2$ (B) Iron from $Fe_2O_3$
(C) aluminium from $Al_2O_3$ (D) magnesium from $MgCO_3.CaCO_3$
**CMT086**

**16.** Upon heating with $Cu_2S$, the reagent(s) that give copper metal is/are **[JEE (Advanced) 2014]**
(A) $CuFeS_2$ (B) CuO (C) $Cu_2O$ (D) $CuSO_4$
**CMT087**

**17.** Copper is purified by electrolytic refining of blister copper. The correct statement(s) about this process is (are) **[JEE (Advanced) 2015]**
(A) Impure Cu strip is used as cathode (B) Acidified aqueous $CuSO_4$ is used as electrolyte
(C) Pure Cu deposits at cathode (D) Impurities settle as anode-mud
**CMT088**

**18.** Match the anionic species given in Column-I that are present in the ore(s) given in Column-II **[JEE (Advanced) 2015]**

| **Column - I** | **Column - II** |
|---|---|
| (A) Carbonate | (P) Siderite |
| (B) Sulphide | (Q) Malachite |
| (C) Hydroxide | (R) Bauxite |
| (D) Oxide | (S) Calamine |
| | (T) Argentite |

(A) (A) - P , R , S ; (B) - R ; (C) - Q , S ; (D) - Q (B) (A) - P , Q , S ; (B) - T ; (C) - Q , R ; (D) - R
(C) (A) - P , S ; (B) – P, T ; (C) - Q , S ; (D) - T (D) (A) - Q , R ; (B) – Q, T ; (C) - Q , T ; (D) – Q
**CMT089**



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**19.** Extraction of copper from copper pyrite ($CuFeS_2$) involves **[JEE (Advanced) 2016]**
(A) crushing followed by concentration of the ore by froth-flotation
(B) removal of iron as slag
(C) self-reduction step to produce 'blister copper' following evolution of $SO_2$
(D) refining of 'blister copper' by carbon reduction

**CMT090**

**20.** Galena (an ore) is partially oxidized by passing air through it at high temperature. After some time, the passage of air is stopped, but the heating is continued in a closed furnace such that the contents undergo self-reduction. The weight (in kg) of Pb produced per kg of $O_2$ consumed is Medium.
(Atomic weights in g mol : O = 16, S = 32, Pb = 207) **[JEE (Advanced) 2018]**

**CMT091**

**21.** Calamine, malachite, magnetite and cryolite, respectively are **[JEE (Advanced) 2019]**
(A) $ZnSO_4$, $CuCO_3$, $Fe_2O_3$ , $AlF_3$
(B) $ZnCO_3$, $CuCO_3.Cu(OH)_2$, $Fe_3O_4$, $Na_3AlF_6$
(C) $ZnSO_4$, $Cu(OH)_2$, $Fe_3O_4$, $Na_3AlF_6$
(D) $ZnCO_3$, $CuCO_3$, $Fe_2O_3$, $Na_3AlF_6$

**CMT092**

**22.** The cyanide process of gold extraction involves leaching out gold from its ore with $CN^{\ominus}$ in the presence of **Q** in water to form **R**. Subsequently, **R** is treated with **T** to obtain Au and **Z**. Choose the correct option(s). **[JEE (Advanced) 2019]**
(A) **T** is Zn (B) **R** is $[Au(CN)_4]^{\ominus}$ (C) **Z** is $[Zn(CN)_4]^{2\ominus}$ (D) **Q** is $O_2$

**CMT093**

**23.** Which among the following statement(s) is(are) true for the extraction of aluminium from bauxite? **[JEE (Advanced) 2020]**
(A) Hydrated $Al_2O_3$ precipitates, when $CO_2$ is bubbled through a solution of sodium aluminate.
(B) Addition of $Na_3AlF_6$ lowers the melting point of alumina.
(C) $CO_2$ is evolved at the anode during electrolysis.
(D) The cathode is a steel vessel with a lining of carbon.

**CMT094**

**24.** The correct statement(s) related to the metal extraction processes is(are) **[JEE (Advanced) 2021]**
(A) A mixture of PbS and PbO undergoes self-reduction to produce Pb and $SO_2$.
(B) In the extraction process of copper from copper pyrites, silica is added to produce copper silicate.
(C) Partial oxidation of sulphide ore of copper by roasting, followed by self-reduction produces blister copper.
(D) In cyanide process, zinc powder is utilized to precipitate gold from $Na[Au(CN)_2]$

**CMT097**

**25.** The electrochemical extraction of aluminum from bauxite ore involves. **[JEE (Advanced) 2022]**
(A) the reaction of $Al_2O_3$ with coke (C) at a temperature $> 2500°C$.
(B) the neutralization of aluminate solution by passing $CO_2$ gas to precipitate hydrated alumina ($Al_2O_3.3H_2O$)
(C) the dissolution of $Al_2O_3$ in hot aqueous NaOH.
(D) the electrolysis of $Al_2O_3$ mixed with $Na_3AlF_6$ to give Al and $CO_2$.

**CMT098**

**26.** The correct option(s) related to the extraction of iron from its ore in the blast furnace operating in the temperature range 900 – 1500 K is(are) **[JEE (Advanced) 2022]**
(A) Limestone is used to remove silicate impurity.
(B) Pig iron obtained from blast furnace contains about 4% carbon.
(C) Coke (C) converts $CO_2$ to CO.
(D) Exhaust gases consist of $NO_2$ and CO. **CMT099**



For More Join: @IITJEE_Advanced

# JEE (Advanced) Practice Paper

This paper is for yourself practice and assessment the discussion of this paper is optional though you can see PDF solutions or video solutions or solutions in hardcopy whichever is provided.

## SECTION–I

- This section contains **TWO** question.
- Each question has FOUR options (A), (B), (C) and (D). ONLY ONE of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories:

Full Marks : +3, if only the bubble corresponding to the correct option is darkened.
Zero Marks : 0, if none of the bubbles is darkened.
*Negative Marks* : –1, in all other cases

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**1.** Metals and their methods of refinement are given
(i) Ni, Cu, Zr, Ga
(ii) Electrolysis, van-Arkel process, Zone refining, Mond's process
Choose the right method for each.
(A) Ni : Electrolysis, Cu : van-Arkel process, Zr : Zone refining, Ga : Mond's process
(B) Ni : Mond's process, Cu : Electrolysis, Zr : van-Arkel process, Ga : Zone refining
(C) Ni : Mond's process, Cu : van-Arkel process, Zr : Zone refining, Ga : Electrolysis
(D) Ni : Electrolysis, Cu : Zone refining, Zr : van-Arkel process, Ga : Mond's process

**CMT052**

**2.** Are electrolytic method of reduction is employed for the preparation of metals that
(A) Are weakly electropositive (B) Are moderately electropositive
(C) Are strongly electropositive (D) From acidic oxides

**CMT053**

## SECTION-II

- This section contains **SEVEN** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4, if only (all) the correct option(s) is (are) chosen.
Partial Marks : +3, if all the four options are correct but ONLY three options are chosen.
Partial Marks : +2, if three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1, if two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0, if none of the options is chosen (i.e. the question is unanswered).
Negative Marks : –2 in all other cases.

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For More Join: @IITJEE_Advanced

**3.** The method of electrolytic refining is suitable in the extraction of :

(A) Aluminium (B) Magnesium (C) Mercury (D) Calcium

**CMT054**

**4.** During the process of electrorefining of copper some metals present as impurity settle as anode mud. These are:

(A) Sn (B) Pb (C) Ag (D) Au

**CMT055**

**5.** In the manufacturing of metallic sodium by fused salt-electrolysis method (Down's process), small amount of $CaCl_2$ that added is known as auxiliary electrolyte and is used to

(A) improve the electrical conductance (B) decrease the melting point of electrolyte

(C) stabilise the metallic sodium (D) Increase the temperature of electrolysis

**CMT056**

**6.** Hoop's process of purification of aluminium involves formation of layers during electrolysis. It involves

(A) the three layers have same densities but different materials.

(B) the three layers have different densities

(C) the upper layer is of pure aluminium which acts as a cathode

(D) the bottom layer is of impure aluminium which acts as an anode and middle layer consists of cryolite and $BaF_2$.

**CMT057**

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**7.** Metallurgical process of zinc involves roasting of zinc sulphide followed by reduction. Metallic zinc distills over as it is volatile and impurities like Cu, Pb and Fe gets condensed. The crude metal obtained is called spelter, which may be purified by

(A) electrolysis process (B) fractional distillation

(C) polling (D) heating with iodine

**CMT058**

**8.** During extraction of copper, it is obtained in the form of molten matte. Which of the following is **not true**?

(A) matte is further treated in Bessemer's converter

(B) molten matte is electrolysed

(C) It is treated with a blast of air and sand

(D) It is dissolved in $CuSiF_6$ and crystallised.

**CMT059**

**9.** The major role of fluorspar ($CaF_2$) which is added in small quantities in the electrolytic reduction of alumina dissolved in fused cryolite ($Na_3AlF_6$) is

(A) as a catalyst

(B) to make the fused mixture very conducting

(C) to lower the melting temperature of the mixture

(D) to decrease the rate of oxidation of carbon at the anode.

**CMT060**



For More Join: @IITJEE_Advanced

**SECTION–III**

- This section contains **TWO** question.
- Each question has matching lists. The codes for the lists have choices (A), (B), (C) and (D) out of which ONLY ONE is correct
- For each question, marks will be awarded in one of the following categories:

Full Marks : +3, if only the bubble corresponding to the correct option is darkened.
Zero Marks : 0, if none of the bubbles is darkened.
Negative Marks : –1, in all other cases.

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**10.** Match column (I) (process) with column (II) (electrolyte)

| **Column (I) (process)** | **Column (II) (electrolyte)** |
|---|---|
| (A) Downs cell | (P) fused $MgCl_2$ |
| (B) Dow's sea water process | (Q) fused ($Al_2O_3$ + $Na_3AlF_6$ + $CaF_2$) |
| (C) Hall-Heroult | (R) fused (40% NaCl + 60% $CaCl_2$) |
| | (S) ($A\ell N$ + C + $N_2$) |

(A) (A) Q ; (B) P; (C) R
(B) (A) P ; (B) P; (C) P
(C) (A) R ; (B) P; (C) Q
(D) (A) Q ; (B) P; (C) R

**CMT061**

**11.** Match column - I with column - II

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| **Column - I (Property)** | **Column - II (Element/compound)** |
|---|---|
| (A) Explosive | (P) Cu |
| (B) Self-reduction | (Q) $Fe_3O_4$ |
| (C) Ferrimagnetic material | (R) $Cu(CH_3COO)_2.Cu(OH)_2$ |
| (D) Verdigris | (S) $Pb(NO_3)_2$ |

(A) (A) S ; (B) P; (C) Q; (D) R
(B) (A) Q ; (B) P; (C) S; (D) R
(C) (A) R ; (B) S; (C) Q; (D) R
(D) (A) Q ; (B) P; (C) S; (D) R

**CMT062**

**SECTION–IV**

- This section contains **ONE** questions of matching type.
- This section contains **One** table (having 3 columns and 4 rows)
- Based on table, there are **THREE** questions
- Each question has FOUR options (A), (B), (C) and (D). ONLY ONE of these four options is correct
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3, if only the bubble corresponding to the correct option is darkened.
Zero Marks : 0, if none of the bubbles is darkened.
Negative Marks : –1, in all other cases.

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For More Join: @IITJEE_Advanced

**Answer Q.12, Q.13 and Q.14 by appropriately matching the information given in the three columns of the following table.**

| Column - I<br>Extraction of metal | Column - II<br>Methods for Reduction | Column - III<br>Refining Method |
|---|---|---|
| (1) $SnO_2 \rightarrow Sn$ | (i) Carbon Reduction | (P) Poling |
| (2) $Al_2O_3 \rightarrow Al$ | (ii) Hydro metallurgic Reduction | (Q) Electrolytic Refining |
| (3) $Cu_2S \rightarrow Cu$ | (iii) Electrolytic Reduction | (R) Distillation |
| (4) $ZnS \rightarrow Zn$ | (iv) Self-Reduction | (S) Puddling Process |

**12.** Which of the following is **NOT** correctly matched?

(A) (1), (iv), (S) (B) (3), (iv), (P) (C) (4), (i), (R) (D) (2), (iii), (Q)

**CMT063**

**13.** Which of the following match is the **CORRECT** reduction and purification methods for Zn

(A) (i), (R) (B) (iv), (S) (C) (iv), (P) (D) None of these

**CMT064**

**14.** Which of the following set of code shows the **CORRECT** similarity with the extraction processes for Pb?

(A) (1), (ii), (S) (B) (4), (iii), (P) (C) (2), (iii), (Q) (D) (3), (iv), (Q)

**CMT065**

## SECTION–V

- This section contains **TWO** paragraph.
- Based on each paragraph, there are **TWO/THREE** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories:

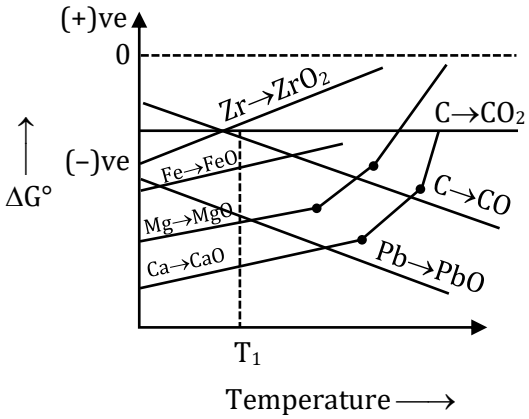
*Full Marks* : +3 if only the bubble corresponding to the correct answer is darkened.

*Zero Marks* : 0 in all other cases.

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### PARAGRAPH FOR QUESTION NO. 15 - 17

### ELLINGHAM DIAGRAM




4



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**15.** Which of the above curve is wrongly presented -

(A) $C \rightarrow CO_2$ (B) $Pb \rightarrow PbO$ (C) $Zr \rightarrow ZrO_2$ (D) $Mg \rightarrow MgO$

**CMT066**

**16.** Which of the above metal oxide is having minimum thermal decomposition temperature.

(A) CaO (B) FeO (C) $ZrO_2$ (D) MgO

**CMT067**

**17.** Which of the following metal's oxide can be reduced by Fe as reducing agent at temperature ($T_1$)

(A) Zr (B) Ca (C) Mg (D) None of these

**CMT068**

**PARAGRAPH FOR QUESTION NO. 18 - 19**
**PURIFICATION METHOD**

At high temperature carbon reacts with water to produce a mixture of carbon monoxide, CO and hydrogen, $H_2$.

$$C + H_2O \xrightarrow{\text{red heat}} CO + H_2$$

CO is separated from $H_2$ and then used to separate nickel from cobalt by forming a volatile compound, nickel tetracarbonyl, $Ni(CO)_4$.

$$Ni + 4CO \longrightarrow Ni(CO)_4$$

**18.** How many moles of $Ni(CO)_4$ could be obtained from the CO produced by the reaction of 75.0g of carbon ? Assume 100% reaction and 100% recovery in both steps.

(A) 6.25 (B) 1.563 (C) 3.125 (D) 25.0

**CMT069**

**19.** Formation of volatile $Ni(CO)_4$ and its subsequent heating gives pure Ni. Process is called -

(A) Hall (B) Dow (C) Serpeck (D) Mond

**CMT070**

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**SECTION–VI**

- This section contains **ONE** question.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3, if only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0, in all other cases.

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**20.** Total number of process which is used for extraction of Aluminium -

Van arkel process, serpeck's process , baeyer's process , hall's heroult's process

**CMT071**



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# ANSWER KEY

## EXERCISE - O

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | C | B | D | C | A | C | B | C | D | B |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | C | C | D | A | A | C | B | D | D | C |
| Que. | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| Ans. | A | A | C | C | B | B | D | B | B | A |
| Que. | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| Ans. | B | D | D | D | D | C | A | B | C | A |
| Que. | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| Ans. | A | A | B | B | C | C | A | C | C | C |
| Que. | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| Ans. | D | C | C | B | B | D | B | A | D | B |
| Que. | 61 | 62 | 63 | 64 | | | 65 | 66 | | |
| Ans. | ABD | B | C | (A) Q; (B) R; (C) S; (D) P | | | A | (A) P; (B) R; (C) Q, R, S; (D) Q, S | | |
| Que. | 67 | | | | 68 | | | | | |
| Ans. | (A) S ; (B) Q ; (C) R ; (D) P | | | | A | | | | | |

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## EXERCISE - S

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | 4 | 2 | 4 | 2 | 4 | 4 | 3 | 2 | 4 | 3 |
| Que. | 11 | 12 | 13 | | | | | | | |
| Ans. | 4 | 4 | 4 | | | | | | | |

## EXERCISE - JEE (Main) PYQ

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | 2 | 4 | 4 | 3 | 4 | 4 | 2 | 4 | 2 | 1 |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | 2 | 4 | 2 | 1 | 1 | 3 | 2 | 4 | 1 | 4 |
| Que. | 21 | | | | | | | | | |
| Ans. | 1 | | | | | | | | | |



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## EXERCISE - JEE (Advanced) PYQ

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | Sintering, Smelting | A | C | B | B | D | B | A | D | C |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | AD or ACD | D | B | A | CD | BCD | BCD | B | ABC | 6.47 |
| Que. | 21 | 22 | 23 | 24 | 25 | 26 | | | | |
| Ans. | B | ACD | ABCD | ACD | BCD | ABC | | | | |

## JEE (Advanced) Practice Paper

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Section-I | Q. | 1 | 2 | | | | | | |
| | A. | B | C | | | | | | |
| Section-II | Q. | 3 | 4 | 5 | 6 | 7 | 8 | 9 | |
| | A. | ABD | CD | AB | BCD | AB | BD | BC | |
| Section-III | Q. | 10 | 11 | | | | | | |
| | A. | C | A | | | | | | |
| Section-IV | Q. | 12 | 13 | 14 | | | | | |
| | A. | A | A | D | | | | | |
| Section-V | Q. | 15 | 16 | 17 | 18 | 19 | | | |
| | A. | B | C | A | B | D | | | |
| Section-VI | Q. | 20 | | | | | | | |
| | A. | 3 | | | | | | | |

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